Reconsidering a Career Decision and an IT related Problem

Decision: Leaving Football for an Engineering Degree

What I did at the time

At 20, after finishing high school, I had to choose between two futures. The first was to continue pursuing football while enrolling in a remote degree that did not require attendance. I had trained and competed for years, and my coach believed I might progress if I committed fully. The second was to stop competitive football and attend an IT/engineering college whose laboratories and seminars required full physical attendance. The timetables made combining them unrealistic.

I did not use a formal decision process. I spoke separately with my coach, parents, and two friends. My coach emphasised that football careers are short and that leaving then would probably close the door permanently. My parents preferred college because it offered a recognised qualification and more stable employment. I searched online for junior IT salaries and looked at several graduate job advertisements, but I did not investigate the remote course carefully or obtain objective information about my probability of progressing in football.

For about two weeks, I repeatedly imagined the two options. Football evoked excitement, belonging, and fear of later regretting that I had abandoned a talent. The engineering college felt less exciting but safer. I listed informal advantages and disadvantages in my head: enjoyment and an uncertain sporting opportunity on one side; employability, intellectual interest, and security on the other. After one poor match in which I was left on the bench, I became doubtful about my sporting prospects. That evening I told my parents I would go for the engineering degree. I informed my coach the following day. My actual rule was essentially: choose the route that seems more secure and is least likely to leave me without a career.

Rewinding the decision with module knowledge

If I could repeat the process, I would first use Halpern and Dunn's (2023) critical-thinking model: identify the goal, gather and evaluate relevant information, consider alternatives and consequences, and monitor the reasoning rather than accepting my first impression. My goal would be "choose the route most likely to produce a satisfying, sustainable life over ten years," not merely "avoid unemployment." I would add two genuine alternatives: defer college for one football season with a pre-agreed review point, and ask whether a lower league training schedule could coexist with the engineering course.

I would then apply expected utility rather than merely mention it. I would score outcomes from 0 (very poor) to 100 (excellent), using information available at 20 rather than what I now know. For engineering, I would estimate a .75 probability of graduating and entering IT with utility 85, a .20 probability of graduating but disliking the career with utility 45, and a .05 probability of dropping out with utility 20. Its expected utility would be $(.75 \times 85) + (.20 \times 45) + (.05 \times 20) = 73.75$. For football plus remote college, I would estimate a .10 probability of becoming a well-paid professional (95), a .35 probability of playing semi-professionally while completing the degree slowly (60), and a .55 probability of neither establishing a football career nor gaining a strong qualification (25). Its expected utility would be 44.25. A one-year deferral might yield .20 × 90 for meaningful football progress, .55 × 65 for returning to engineering one year later, and .25 × 30 for injury or educational drift, giving 61.25. These are subjective estimates, but writing them down would expose assumptions for challenge. I would ask my coach for progression statistics for comparable players, speak to current students from both courses, and revise the probabilities.

I would also regulate “hot affect”. Emotions can shape risk perception, information processing, and choices, including through incidental feelings not caused by the decision itself (Lerner et al., 2015). Therefore, I would not decide immediately after being benched. I would calculate the options twice, one week apart, and record my emotional state. Prospect theory also predicts loss aversion: losses can loom larger than equivalent gains (Kahneman & Tversky, 1979). I framed uncertain football as a threat of “losing” a safe career, while my coach framed college as permanently “losing” football. I would neutralise this by describing every option in both gain and loss frames and asking two independent people to critique the table.

Comparison and reflection

My naive approach considered relevant values, but it relied on vivid advice, one bad match, and an undefined preference for safety. The module-informed approach makes alternatives, probabilities, values, emotions, and framing visible. It still would not guarantee the best outcome: utilities combine identity, money, health, and relationships imperfectly, while probabilities about a unique sporting career are uncertain. Moreover, hindsight bias could make engineering appear obviously correct because my subsequent IT career went well. Nevertheless, the structured method would have increased the chance of a defensible decision. Interestingly, it probably produces the same choice, but for transparent reasons rather than because disappointment after one match tipped the balance.

Problem: Intermittent Failures in a Software Release

What I did at the time

Five years into my IT career, I was responsible for a payment service after a Friday release. Some customers payments failed, yet retries often succeeded. Monitoring showed database time-outs, so I immediately assumed the database was overloaded. A senior manager wanted an answer within two hours. I restarted the database connection pool and increased its capacity. Failures briefly decreased, then returned.

I searched for recent code changes and asked two developers to inspect database queries. We found one slow query and added an index, but the error rate remained unstable. I then rolled back the whole release. This stopped most failures, although several continued, which I attributed to transactions already in progress. On Monday, a network engineer found that a new firewall rule was intermittently dropping connections between two application servers and the database. The release increased traffic and exposed the fault, but its code was not the primary cause. We corrected the rule and redeployed successfully. My response restored service, but I had spent hours pursuing the first plausible explanation and could not initially explain why the rollback worked only partially.

Rewinding the problem with module knowledge

I would represent the incident as a problem space: an initial state, a goal state, permissible operators, and constraints (Newell & Simon, 1972). The initial state would be “6% intermittent payment failures after deployment; database time-outs present; cause unknown”. The goal state would be “failures below 0.1% for 30 minutes, no lost transactions, and evidence identifying the cause”. Constraints would include customer harm, a two hour restoration target, incomplete logs, data integrity requirements, and only four available staff. Operators would include comparing healthy and failing requests, querying logs, isolating one server, reverting one change, testing connectivity, increasing capacity, or rolling back.

Using means-ends analysis, I would divide the gap into subgoals. First, contain harm: pause non-essential batch traffic and enable safe retries. Second, localise the fault: create a table by application server, time, transaction type, and database node. Third, discriminate among hypotheses. H1 would be database overload, H2 faulty release code, and H3 network or firewall failure. Database CPU below 45% would weaken H1; failures concentrated on two servers would strengthen H3; failures across all servers only on the new version would strengthen H2. Fourth, implement the smallest reversible fix, then verify the 0.1% criterion.

Instead of changing several variables, I would run a low risk causal test. I would route 10% of traffic from one affected server through a known-good network path while holding the software version constant. If its failure rate fell while a matched server remained unstable, that result would discriminate network failure from defective code. I would then test the firewall configuration and roll back only the rule. A visual diagram of application-to-database routes would also restructure my representation: “database time-out” describes where the symptom appeared, not necessarily where the cause originated. Externalising the topology could reveal the shared network component that my code-centred mental model excluded (Halpern & Dunn, 2023).

If evidence remained ambiguous after containment and initial tests, I would record the hypotheses and take a short, deliberate break before reviewing them with a colleague. Incubation means setting a problem aside after preparation; meta-analytic evidence indicates a positive overall effect, although its benefit depends on task and incubation conditions (Sio & Ormerod, 2009). Under an active outage, a ten minutes handover is more realistic than sleeping on the problem.

Comparison and reflection

The naive process was a loosely guided search: I anchored on the first salient symptom, altered capacity, changed a query, and finally used the broad operator of full rollback. The informed process narrows the search through explicit states, constraints, subgoals, and diagnostic interventions. It would probably have located the firewall sooner while preserving a rapid rollback as a satisficing containment option. However, a formal map costs time and evolving incidents can change the problem space. The sensible standard is therefore not a perfect explanation before acting, but fast containment followed by controlled tests. Although luck still affects outcomes, consistently separating symptoms from causes and testing competing explanations would improve the probability of an efficient, well-supported solution.

References

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